Windows SDK: Download and install Windows SDK 7 from the link given in References [2]. Windows SDK provides the tools, compilers, headers and libraries that are necessary to run Hadoop.

Cygwin: Hadoop requires UNIX command line tools like Cygwin or GnuWin32. Download and install Cygwin from the link given in References [3], to its default location *C:\cygwin64* and make sure to select the 'openssh' package and its associated prerequisites from the *Select packages* tab.

Maven and the Protocol buffer: Install Maven 3.0 or later and the Protocol buffer 2.5.0 into the *C:\maven* and *C:\protobuff* directories, respectively.

Setting environment variables

Navigate to *System properties---> Advanced ---> Environment Variables*. Add environment variables *JAVA_HOME, M2_HOME (for Maven)* and *Platform* (x64 or Win32 depending on the system architecture).

Note that the variable name, Platform, is case sensitive and values will be *x64* or *Win32* for 64-bit and 32-bit systems. Edit the path variable under *System variables* to add the following: *C:\cygwin64\bin;C:\cygwin64\usr\sbin;C:\maven\bin; C:\protobuf*

The dos and don'ts

Given below are some of the issues usually faced while installing Hadoop on Windows.

If the JAVA_HOME environment variable is set improperly, Hadoop will not run. Set environment variables properly for JDK, Maven, Cygwin and Protobuffer. If you still get a 'JAVA_HOME not set properly' error, then edit the *C:\hadoop\bin\hadoop-env.cmd* file, locate 'set *JAVA_HOME =*' and provide the JDK path (with no spaces).

Do not use the Hadoop binary, as it is bereft of *Windowsutils.exe* and some *Hadoop.dll* files. Native IO is mandatory on Windows and without it the Hadoop installation will not work on Windows. Instead, build from the source code using Maven. It will download all the required components.

Download Hadoop from the link in References [4]. Building from the source code requires an active Internet connection the first time.

Building and configuring Hadoop on Windows

Select *Start ---> All Programs ---> Microsoft Windows SDK v7.1* and open the Windows SDK 7 command prompt as the administrator. Change the directory to *C:\hadoop* (if it doesn't exist, create it). Run the command prompt:

```
C:\Program Files\Microsoft SDKs\Windows\v7.1>cd c:\hadoop
C:\hadoop>mvn package -Pdist,native-win -DskipTests –Dtar
```

Note: The above step requires an active Internet connection because Maven will download all the dependencies. A successful build will generate a native binary package *C:\hadoop\hadoop-dist\target\hadoop-2.2.0.tar.gz* directory. Extract hadoop-2.2.0.tar.gz under *C:\hdp.*

Starting a single node installation: If the build is successful, check out which Hadoop version by running the following command:

```
C:\hadoop>hadoop version
```

Startup scripts: The *C:\hdp\bin* directory contains the scripts used to launch Hadoop DFS and MapReduce daemons.

start-dfs.cmd: Starts the Hadoop DFS daemons, the namenode and datanode.

start-mapred.cmd: Starts the Hadoop MapReduce daemons, the jobtracker and tasktrackers.

start-all cmd: Starts all Hadoop daemons, the namenode, datanode, the jobtracker and tasktrackers.

start-dfs.cmd: Starts the Hadoop DFS daemons, the namenode and datanode. Use this before *start-mapred.sh*

start-mapred.cmd: Starts the Hadoop MapReduce daemons, the jobtracker and tasktrackers.

start:all.cmd - Starts all Hadoop daemons, the namenode, datanode, the jobtracker and tasktrackers.

The following section contains details on how to configure Hadoop on Windows. Assuming the install directory is *C:\hdp,* run the command *C:\hadoop> cd C:\hdp\etc\hadoop*

Edit the file *hadoop-env.cmd* (which contains the environment variable settings used by Hadoop) in Notepad++ and add the following lines at the end of the file. Run the command prompt:

```
set HADOOP_PREFIX=c:\hdp
set HADOOP_CONF_DIR=C:\hdp\etc\hdp
set YARN_CONF_DIR=C:\hdp\etc\hadoop
set PATH=%PATH%;C:\hdp\bin
```

Before we get started with setting Hadoop environment variables and running Hadoop daemons, we need to configure the following files: *core-site.xml, hdfs-site.xml, yarn-site.xml mapred-site.xml* and the slave files located in *C:\hdp\etc\hadoop*. The minimum configuration settings are given below.

Edit or create the file *C:\hdp\etc\hadoop\core-site.xml* (all Hadoop services and clients use this file to locate namenode; it contains the name of the default file system) and make sure it has the following configuration key:

```
<configuration>
<property>
<name>fs.defaultFS</name>
<value>hdfs://localhost:9000</value>
</property>
</configuration>
```

Edit or create the file *C:\hdp\etc\hadoop\hdfs-site.xml* (HDFS services use this file and it contains HTTP addresses for namenode and datanode) and add the following configuration key: